

KALKAR, Germany

WMO: 104040

Lat: 51.750N

Lon: 6.283E

Elev: 43

StdP: 100.81

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-5.2	-10.6	1.5	-6.3	-8.3	1.9	-4.2	11.6	8.8	10.1	8.7	2.2	70	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	30.4	20.0	28.1	19.1	26.0	18.1	21.0	28.0	20.0	26.6	19.0	24.7	2.4	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.5	23.8	17.6	12.7	22.6	16.7	11.9	21.8	60.8	27.9	57.2	26.5	53.9	24.7	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.1	6.0	DB	-10.3	34.4	3.3	1.4	-12.7	35.5	-14.6	36.3	-16.5	37.1	-18.9	38.2	(4)
(5)			WB	-10.6	23.4	3.3	0.8	-12.9	24.0	-14.8	24.5	-16.6	25.0	-19.0	25.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	2.8	4.0	6.6	10.3	14.0	16.7	18.7	18.7	15.4	11.3	6.9	3.1
	DBStd	6.80	4.46	4.13	3.56	3.74	3.59	3.38	3.27	3.07	2.89	3.37	3.51	4.25
	HDD10.0	899	224	169	114	41	7	0	0	0	1	26	102	216
	HDD18.3	2894	482	401	363	241	139	71	34	32	96	219	343	473
	CDD10.0	1177	1	2	9	51	132	202	271	271	163	65	9	2
	CDD18.3	130	0	0	0	1	7	23	47	44	9	0	0	0
	CDH23.3	1182	0	0	0	19	86	219	426	368	62	1	0	0
	CDH26.7	362	0	0	0	2	16	62	146	126	11	0	0	0
Wind	WSAvg	2.6	3.3	3.3	2.9	2.5	2.3	2.1	2.2	2.1	2.2	2.6	2.8	3.1
Precipitation	PrecAvg	827	71	63	55	44	63	74	81	86	69	68	75	79
	PrecMax	1064	131	130	123	88	135	181	154	215	167	205	117	164
	PrecMin	641	4	14	10	0	12	10	20	24	9	21	8	26
	PrecStd	104	34	31	32	21	28	40	35	50	41	33	32	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	14.0	18.9	26.0	28.9	31.8	33.3	33.3	28.3	22.1	16.1	13.2
		MCWB	10.5	9.9	12.1	14.7	18.7	21.0	20.8	21.1	19.1	16.9	12.9	11.3
	2%	DB	11.3	12.1	16.1	22.3	25.7	28.2	30.4	30.0	25.1	19.3	14.2	11.8
		MCWB	9.5	9.3	11.0	13.9	16.1	19.2	19.9	20.0	18.5	15.4	12.1	10.2
	5%	DB	10.1	10.9	13.9	19.6	23.5	25.8	27.9	27.4	22.4	17.5	13.0	10.3
		MCWB	8.3	8.6	9.9	12.5	15.2	18.1	19.2	19.1	16.9	14.4	11.3	8.8
	10%	DB	8.7	9.6	12.0	17.1	21.1	23.4	25.3	24.8	20.3	16.0	11.7	9.1
		MCWB	7.2	7.7	9.0	11.5	14.4	16.8	18.4	18.2	15.7	13.6	10.0	7.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	10.9	13.0	16.2	19.6	22.0	22.6	22.4	20.3	17.4	13.5	11.7
		MCDB	12.2	12.5	17.1	24.2	26.8	29.5	30.2	30.5	25.9	20.9	15.2	12.9
	2%	WB	9.7	9.8	11.7	14.5	17.7	20.1	20.9	20.9	18.7	16.0	12.5	10.4
		MCDB	11.2	11.5	15.0	20.6	23.6	26.8	27.6	28.1	23.7	18.8	14.1	11.6
	5%	WB	8.4	8.8	10.4	13.1	16.2	18.7	19.9	19.7	17.5	14.9	11.4	9.0
		MCDB	9.9	10.6	13.2	18.5	21.2	24.3	26.5	25.9	21.6	17.0	12.9	10.2
	10%	WB	7.1	7.8	9.2	11.9	15.0	17.5	18.8	18.6	16.3	13.8	10.2	7.7
		MCDB	8.5	9.5	11.8	16.6	20.0	22.3	24.5	23.8	19.7	15.8	11.4	9.1
Mean Daily Temperature Range		MDBR	4.7	5.6	7.5	9.8	9.9	10.1	9.7	9.9	8.7	7.3	5.3	4.4
	5% DB	MCDBR	5.3	6.8	10.7	13.8	14.1	14.0	13.8	14.1	11.6	8.7	6.4	5.0
		MCWBR	4.5	5.1	6.8	7.3	7.1	7.1	6.2	6.5	6.5	5.9	5.1	4.5
	5% WB	MCDBR	5.2	5.8	9.3	12.2	12.2	12.1	12.4	12.3	10.1	8.1	6.0	4.9
		MCWBR	4.6	4.7	6.3	7.0	7.1	6.9	6.2	6.4	6.2	6.0	5.1	4.7
Clear-Sky Solar Irradiance	taub	0.330	0.353	0.382	0.416	0.420	0.418	0.429	0.420	0.394	0.366	0.344	0.325	
	taud	2.362	2.334	2.256	2.199	2.227	2.250	2.226	2.273	2.335	2.394	2.370	2.338	
	Ebn at Noon	668	757	802	818	834	838	820	806	784	732	643	603	
	Edh at Noon	63	84	109	130	132	130	131	119	100	79	62	55	
All-Sky Solar Radiation	RadAvg	0.72	1.39	2.52	4.07	4.94	5.29	5.04	4.27	3.11	1.83	0.85	0.55	
	RadStd	0.10	0.26	0.36	0.45	0.43	0.44	0.62	0.38	0.35	0.16	0.10	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (84 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page